LESSON PLAN

AIRCRAFT ARMAMENT (18)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	01
Total No. of Lesson Plans	12	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Introduction to surveillance system Surveillance systems of the aircrafts.
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Introduction to surveillance system Surveillance systems of the aircrafts.
5. Specific Objectives	18.1.3.1 Trainee will be able to describe the basic theory of operation of surveillance system of an aircraft.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Review of IB diagrams regarding surveillance loops; Instructor supervision required for diagram interpretation.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the primary purpose of an aircraft surveillance system?
(b) List three types of aircraft surveillance systems.
(c) Why is surveillance critical for air traffic control?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Definition of aircraft surveillance • Importance of surveillance in aviation • Basic components of surveillance • Types of surveillance systems • General theory of operation	L & Ds	PPT, IB	15 min
(b)	• Active vs passive surveillance • Radar-based detection principles • Signal transmission and reception • Data processing and display • System integration with avionics	L & Ds	PPT, PDF	20 min
(c)	• Surveillance system limitations • Common failure modes • Maintenance considerations • Safety implications of failure • Future trends in surveillance	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain the basic theory of operation for a standard aircraft surveillance system.
(b) Differentiate between active and passive surveillance methods.

12. Home Assignment and Follow Up	Prepare a brief note on how the principles of surveillance discussed in this lesson apply to the SSR/transponder system covered in LP 02.

Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Introduction to SSR/transponder System. Interrogators responders. Main technological characteristics of the System.	Refer to the master syllabus reference list	LP 02

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min